MOTOR DURUM IZLEME KARTI

WEG W22 IE3 15 kW referans motoru — ariza agaci analizinden turetilmistir

Besleme 24 V DC — sebeke yalnizca role kontaginda

01—guc

Guc — 24 V giris, 3,3 V buck, 3,0 V analog

Dosya: 01—guc.kicad_sch

02—mcu

MCU — STM32G474 cekirdek

Dosya: 02—mcu.kicad_sch

03—akim

Akim on ucu — 3 x CT

Dosya: 03—akim.kicad_sch

04—ptc

PTC izleme — izole on uc

Dosya: 04—ptc.kicad_sch

05—role

Role cikisi ve yalitim bariyeri

Dosya: 05—role.kicad_sch

06—arayuz

LCD, butonlar, LED, flash

Dosya: 06—arayuz.kicad_sch

07—haberlesme

RS—485 x 2 — sensor basligi + Modbus

Dosya: 07—haberlesme.kicad_sch

Melike Atılgan

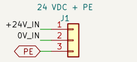
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Title: Motor Durum Izleme Karti

Size: A3	Date: 2026—10—11	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 1/8

GIRIS KORUMA

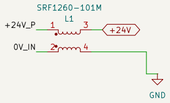


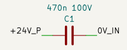


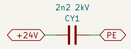


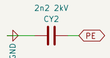


ORTAK MOD FILTRE









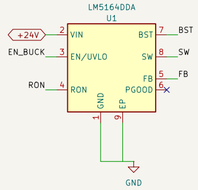


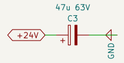


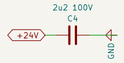


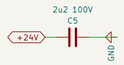


24 V –> 3,3 V SENKRON BUCK





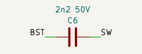




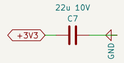


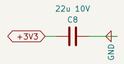


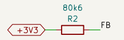












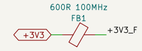


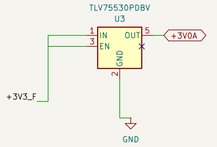




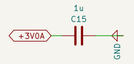


ANALOG BESLEME 3,0 V









TASARIM NOTLARI

1) Karta sebeke gerilimi getirilmez. Besleme 24 V DC pano barasi (IEC 61131-2: 19,2–30 V). Sebeke yalnizca 05-role sayfasindaki role kontagindadir.
2) GIRIS KORUMASI. D1 SMBJ33CA cift yonlu: 33 V a kadar iletmez, darbede en cok 53,3 V ta kirpar. D2 seri Schottky ters baglantiyi keser. Negatif darbede D2 uzerindeki ters gerilim 53 + 30 = 83 V olabilir; bu yuzden 100 V luk diyot secildi. U1 in giris siniri 100 V.
3) L1 SRF1260-101M: sargilar 1–3 ve 2–4 bacaklari arasinda (uretici foyu). Hat 1 den girer 3 ten cikar; donus 4 ten girer 2 den cikar. 101M: seri DCR 0,66 R, Irms 0,89 A. 102M in DCR si 6,8 R idi (0,2 A de 1,4 V).
4) BUCK (LM5164, sabit acik kalma sureli). Vout = (VFB + dV/2) x (1 + R2/R3) R2 = 80k6, R3 = 49k9 –> 3,16 V. En kotu durum (VFB 1,181–1,218 V, %1 direnc): 3,08–3,25 V. LCD (3,0–3,3 V) ve ADM2582E (3,0–3,6 V) icinde. Frekans: Rron(k) = Vout x 2500 / Fsw(kHz) –> 27k4 ile 289 kHz Bobin 33 uH: 24 V ta akim dalgalanmasi 0,29 A; doyma akimi 2,5 A, akim siniri 1,5 A. C6 2,2 nF olmak zorunda (foy: 1,5–2,5 nF).
5) UVLO: acma = 1,5 V x (1 + R4/R5) = 16,9 V, kapama = 1,4 V x (...) = 15,7 V
6) ANALOG BESLEME. U3, 3,3 V tan 3,0 V uretir (ferrit + LDO). ADC referansi ve akim on ucu bu raydan beslenir; amac buck dalgalanmasinin olcume etkisini azaltmak. Orta nokta 1,5 V.
7) TOPRAK: tek GND. Analog/dijital ayrimi duzlem bolerek degil, yerlesimde bolge ayirarak yapilir.
8) CY1/CY2 2,2 nF 2 kV: 24 V devresi ile PE arasinda. Devre SELV oldugu icin Y sinifi kondansator gerekmez; darbe dayanimi icin 2 kV secildi.
Tip 3 ag (R7, C9, C10): FB de 24 V ta 20 mV, 19,2 V ta 19 mV dalgalanma.

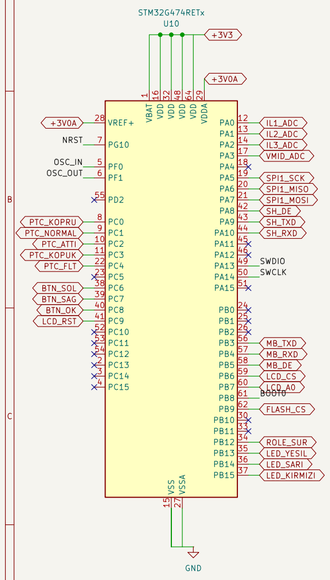
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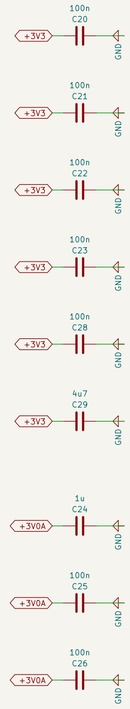
Title: Guc katmani – 24 V DC giris, 3,3 V buck, 3,0 V analog

Size: A2	Date: 2026-10-11	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 2/8

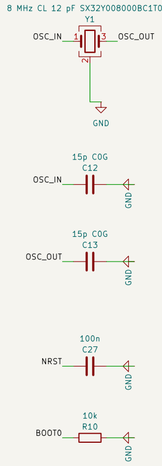
MIKRODENETLEYICI



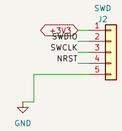
BESLEME AYRISTIRMA



KRISTALLER VE RESET



SWD PROGRAMLAMA



TASARIM NOTLARI

1) VDDA ve VREF+ ayri analog raydan (+3V0A) beslenir. ADC referansi buradan turedigi icin buck gurultusu dogrudan olcume girerdi.

2) Her VDD pinine ayri 100n; VDDA icin 1u + 100n + VREF+ icin 100n.
   Kondansatorler yerlesimde pinin 2 mm yakinina konacak.

3) PA0–PA3 = ADC1_IN1..IN4. Akim kanallari ADC1 uzerinde toplandi;
   kanallarin ornekleme sirasi firmware de ayarlanacak.

4) PB8 = BOOT0. 10k ile asagi cekildi; aksi halde acilista bootloader a
   dusme riski var (G4 te BOOT0 ayri pin degil).

5) NRST te 100n var, pull-up YOK: STM32 de dahili pull-up bulunuyor.

6) Y1 8 MHz 3225, CL 12 pF, ESR <= 300 R (TKD SX32Y008000BC1T001).
   C12 = C13 = 2 x (12 pF – ~4 pF kacak) = 16 -> 15 pF C0G.
   gm_krit = 4 ESR (2 pi f)^2 (C0 + CL)^2 = 0,68 mA/V < 1,5 mA/V (HSE Gmcritmax).
   32,768 kHz kristal yok: kartta yedek pil olmadigi icin saat besleme
   kesilince zaten sifirlanir; zaman damgasini HSE den turetmeyi planladim.

7) Kullanilmayan pinler bagli-degil isaretlendi; bos birakilmadi.

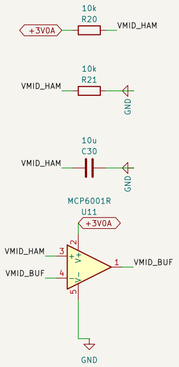
Melike Atılgan

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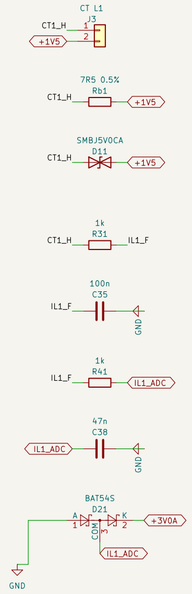
Title: MCU cekirdegi – STM32G474RETx

Size: A2	Date: 2026-10-11	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 3/8

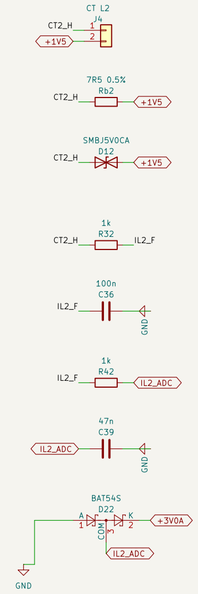
1,5 V ONGERILIM



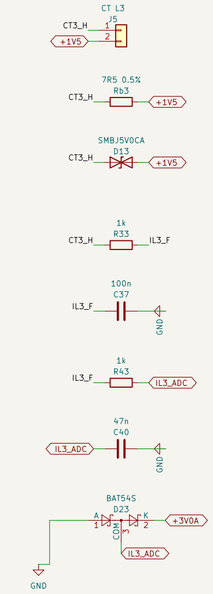
AKIM KANALI L1



AKIM KANALI L2



AKIM KANALI L3



TASARIM NOTLARI

Rb kalicidir, konnektore sokulup cikarilmaz.

1) OLCEKLENDIRME. CT 100 A : 50 mA (2000:1), yuk direnci Rb = 7,5 Ω
   In = 28,3 A -> Is = 14,2 mA -> Vb = 0,106 V RMS (0,150 V tepe)

   Yol verme 198 A -> Is = 99 mA -> Vb = 0,743 V RMS (1,050 V tepe)

   Tam olcek 1,5 V tepe -> 283 A RMS. Kalkis akimi kirpilmaz.
2) ONGERILIM. CT cift yonlu akim veriyor; ADC tek kutuplu. Sinyal

   +1V5 etrafinda salinir. MCP6001 tampon bolucuyu yukten ayirir.

   R23 opamp cikisini 10 uF tan ayirir (kapasitif yuk kararliligi);
   CT akimi tamponun icinden gecmez, kendi yuk direncinde doner.

   +1V5 ayrica VMID_ADC ile MCU tarafindan izlenir.

3) ANTI-ALIAS. Iki kademe RC: 1k + 100n ve 1k + 47n.
   Birinci kademe fc = 1/(2*pi*1k*100n) = 1,59 kHz; ornekleme 4 kHz.

4) KORUMA. Giriste cift yonlu TVS (SMBJ5V0CA): ADC girisinde BAT54S

   cift Schottky GND ve +3V0A ya kirpar. CT ikincil ucu acik kalmamali:

5) KAYALITIM. CT kendi yalitimini saglar; sebeke bu sayfada yok.

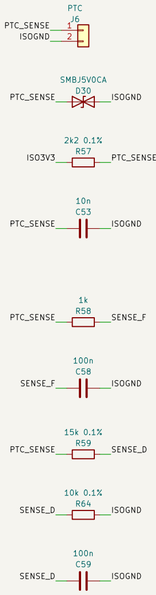
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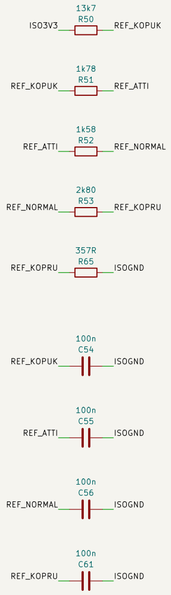
Title: Akim on ucu – 3 x ayrik nuve CT

Size: A2	Date: 2026-10-11	Rev: C
KiCad E.D.A. 10.0.6		Id: 4/8

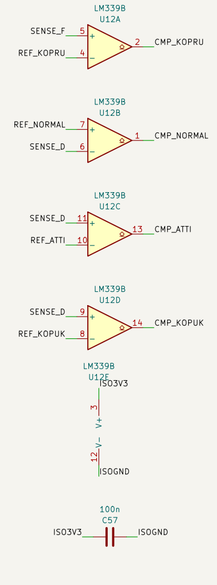
PTC GIRISI (IZOLE TARAF)



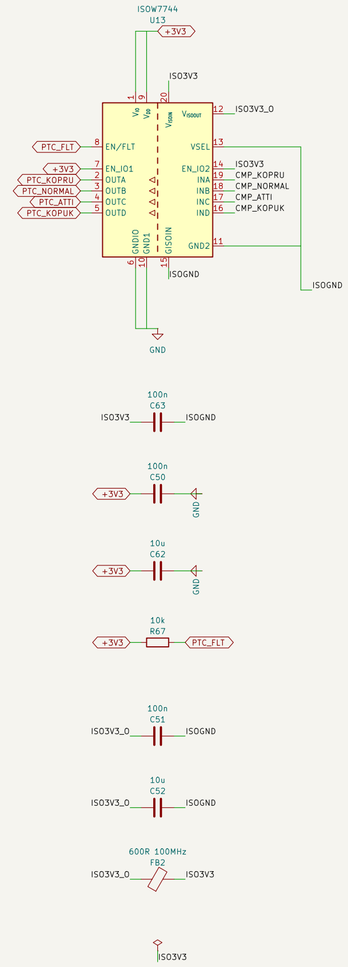
ESIK BOLUCUSU



KARSILASTIRICILAR



IZOLATOR VE IZOLE BESLEME



TASARIM NOTLARI

1) NEDEN IZOLE. PTC sargiya gomuludur; sargi–PTC yalitimi delinirse PTC ucu sebeke potansiyeline cikar. U13 (ISOW7744) takviyeli yalitimlidir:
   yuzeysel kacak ve hava araligi > 8 mm, 5 kV RMS, calisma gerilimi
   1000 V RMS (uretici foyu). Dort veri kanali ve izole 3,3 V ayni kilifta.
2) OLCUM. PTC, R57 (2k2) ile ISO3V3 e cekilir:
   Vsense = ISO3V3 x Rp / (Rp + 2k2). Rp = Rptc // 25k (bolucu yuku)
   Esikler ISO3V3 e oransaldir; izole beslemenin toleransi sonucu etkilemez.
3) ESIKLER (en kotu durum: direnc toleransi, ofset, besleme)
   R < 40 R KOPRU (uc PTC nin soguk direnci en az 60 R)
   R < 1,5 k NORMAL (geri donus esigi; IEC 60947-8: 750–1650 R)
   R > 3,6 k ATTI (acma esigi; IEC 60947-8: 1650–4000 R)
   R > 15 k KOPUK ya da cok sicak; ikisinde de role birakmali
4) ORTAK MOD. LM339B (–40..+85 C) giris araligi 3,3 V ta 0..1,3 V. Kopru esigi 59 mV (dogrudan); diger esikler 0,4 e bolunmus isarette
   0,52 / 0,78 / 1,07 V. Hepsi aralik icinde.
5) ARIZA DAVRANISI. Izole besleme coker ya da U13 arizalanirsa cikislar
   varsayilan YUKSEK e gider: NORMAL ve ATTI ayni anda yuksek olamaz, firmware bunu ariza saymali. PTC_FLT ayrica DC–DC arizasini bildirir.
6) PTC ye uygulanan gerilim en cok 3,3 V, akim en cok 1,5 mA; sensor
   basina 2,5 V olcme gerilimi sinirinin altinda (soguk durumda).
7) TEK ZINCIR UYARISI. Motordaki PTC zinciri ayni anda sertifikali PTC
   rolesine ve bu karta paralel baglanmaz: iki olcum akimi birbirini bozar.
   Motorda tek zincir varsa sertifikali roleye gider; bu giris ikinci zincir (ya da rolenin yardimci kontagi) icindir.

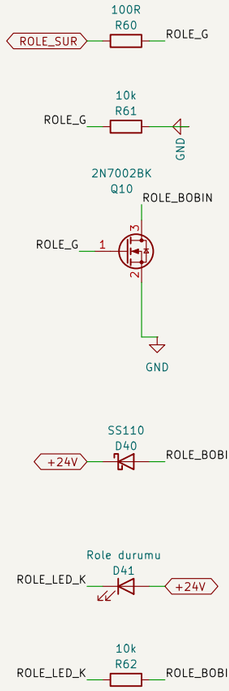
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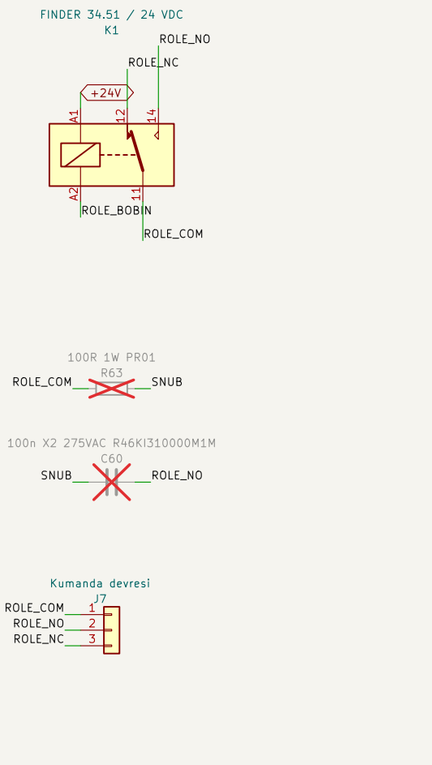
Title: PTC izleme – takviyeli yalitimli on uc

Size: A2	Date: 2026-10-11	Rev: C
KiCad E.D.A. 10.0.6		Id: 5/8

ROLE SURUCUSU (SELV)



ROLE VE SEBEKE BOLGESI



TASARIM NOTLARI

1) KAPALI DEVRE PRENSIBI. K1 in NO kontagi kullaniliyor ve role
NORMAL CALISMADA SUREKLI ENERJILI. Kesme karari veya besleme kaybi
-> role birakir -> kontak ACILIR -> motor durur.
NC kontak kullanilsaydi besleme kaybinda kontak KAPANIR ve kart
devre disi kaldigi halde motor calismaya devam ederdi.
R61 (10k) kapiyi asagi ceker: MCU resetteyken de role birakir.

2) KART ANA KORUMA DEGIL. Bu kontak, kumanda devresine sertifikali
termik role ve PTC rolesiyle SERI girer. Kart arizalansa bile
onlar gorevine devam eder. Ana koruma olmak IEC 60947-4-1 TIP TESTI
ister; bu kart o amacla tasarlanmadi.

3) YALITIM BARIYERI. Sebeke bolgesi = yalnizca K1 in kontak uclari
(11, 14) ve J7 klemensi. IEC 60664-1, 230/400 V, OVC III, PD 2,
malzeme grubu IIIa -> anma darbe dayanimi 4 kV.
Takviyeli yalitim: bosluk >= 5,5 mm yuzeysel kacak >= 8,0 mm
PCB de bariyer K1 in govdesinin altindan gecen FREZELENMIS YARIK;
yarik boyunca hicbir katmanda bakir ve ipek baski yok.

4) AC-15. Kontaktor bobini enduktif AC yuktur. Role kontagi AC-1
degil AC-15 kullanma kategorisinde secilir (IEC 60947-5-1).
R63 + C60 snubber, kontak acilirken olusan gerilim darbesini ve
EMI yi azaltmak icin (su an takilmaz, bkz. 5).
C60 X2 sinifi olmak zorunda (IEC 60384-14).

5) D40 flyback diyot bobine ters paralel: katot +24 V ta, anot Q10 un
drain ucunda. Q10 iletimdeyken ters kutuplanir, kesildiginde bobin
akimini uzerine alir; olmazsa Q10 un drain dayanimi asilir.
Q10 2N7002BK: 60 V, VGSth 1,1-2,1 V. Bobin akimi 7 mA (Finder 34, 24 V).

6) SNUBBER TAKILMAZ (DNP). C60 acik kontak uzerinden 230 V ta ~7 mA
sizdirir; kucuk bir kontaktor bobini bununla BIRAKMAYABILIR ve kart
motoru durduramaz. Kumanda yukunun birakma gerilimi bilinince takilir.
R63 100R 1 W PR01 (aksiyal, 350 V), C60 100 nF X2 275 VAC R46KI310000M1M.

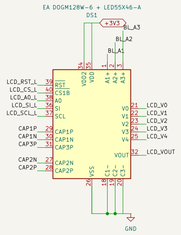
Melike Atılgan

Sheet: /05-role/
File: 05-role.kicad_sch

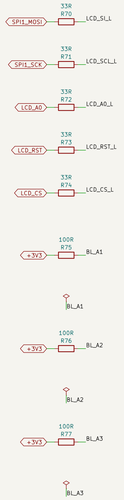
Title: Role cikisi ve yalitim bariyeri

Size: A3	Date: 2026-10-11	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 6/8

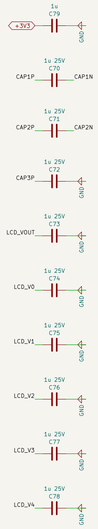
LCD EA DOGM128X-6 (ST7565R)



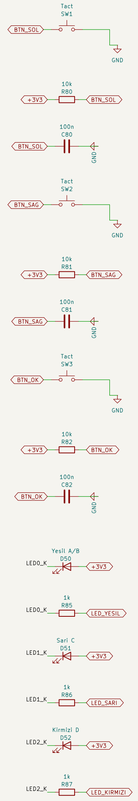
LCD SERI DIRENCLERI (CISPR 11)



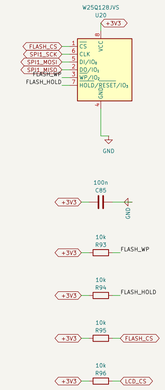
LCD YUK POMPASI



BUTONLAR VE LED



SPI FLASH



TASARIM NOTLARI

1) NEDEN OLED DEGIL. Ekran 7/24 ayni etiketleri ayni yerde gosterecek.
   OLED de statik icerik KALICI IZ (burn-in) birakir; sicaklik araligi
   da dardir. FSTN LCD de burn-in yok, -20...+70 C calisir.
2) SERI 33R. Ekran serit kablosu CISPR 11 Sinif A da isima kaynagidir.
   Dirençler MCU tarafindaki global net ile LCD nin yerel pini ARASINDA
   durur: kenar dikligini yumusatip isimayi dusurur. Yerlesimde LCD
   konnektorunun hemen yanina konacak.
3) BUTONLAR. 10k pull-up + 100n; firmware de ~20 ms debounce eklenecek.
   Sifirlama için OK a 3 s basili tutma dusunuldu (yanlislikla sifirlama).
4) LED renkleri ISO 20816-3 bolgeleri için ayrildi: yesil A/B, sari C, kirmizi D.
   Ortak anot +3V3: MCU pini dusurunce yanar (sink akimi daha kararli).
5) FLASH SPI hatlarinda seri direnc YOK: izler kisa ve kart içinde;
   seri direnc 80 MHz SPI de kenar bozar, isima riski LCD kablosunda.
6) FLASH, kesme anindaki olcumleri zaman damgasiyla saklamak için:
   kesmenin nedeni sonradan kayittan okunabilsin.
7) LCD yuk pompasi kondansatorleri ekranin 2 mm yakinina konacak;
   uzun izler pompa verimini ve kontrasti dusurur.
8) LCD BESLEMESI 3.0–3.3 V: kart rayi bu yuzden 3.16 V (01-guc not 4).
   Pompa 4x (DOGM foyu, 3.3 V uygulamasi): VOUT = 4 x 3.25 = 13.0 V <
   13.5 V. Pompa kondansatorleri 25 V, C79 ekranin VDD bacagina.
9) ARKA ISIK EA LED55x46-A (amber, Vf 1.9 V), uc yol, yol basina 100R:
   (3.16 − 1.9) / 100 = 12.6 mA, toplam 38 mA (foy: en cok 40 mA).

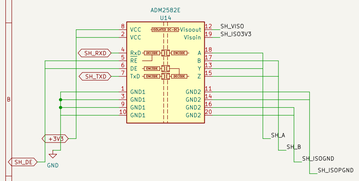
Melike Ahlgan

Sheet: /06-arayuz/
File: 06-arayuz.kicad_sch

Title: Arayuz — LCD, butonlar, LED, SPI flash

Size: A1	Date: 2026-10-11	Rev: C
KiCad E.D.A. 10.0.6		Id: 7/8

SENSOR BASLIGI HATTI



























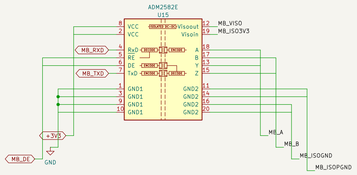








MODBUS RTU (SCADA)

















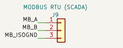


















TASARIM NOTLARI

1) IKI HAT BIRBIRINDEN DE IZOLE. ADM2582E kendi izole beslemesini
   uretir (dahili DC/DC). Sensor basligi hatti motor govdesine,
   Modbus hatti pano/SCADA topragina referanslidir; ikisi arasinda
   potansiyel farki olabilir. Ortak toprak kullanilsaydi toprak
   dongusu olusur, olcum gurultulu olurdu.
2) -RE ve DE AYNI PINE baglandi: yarim cift yonlu (half-duplex)
   otomatik yon kontrolu. MCU bir pin ile surucuyu acip kapatir.
3) SONLANDIRMA 120R. Iki hat da noktadan noktaya (sensor basligi ve
   gecit): arta kart iki hattin da bir ucu, sonlandirma kalici takili.
   Hatta araya baska cihaz eklenirse aradakinde direnc sokulur.
4) SM712 (SOT-23, uc bacak): iki hatti tek klipte korur. SINIR: 1 A
   darbede +19 / -11 V a kirpar. ADM2582E hat bacaklari -9 / +14 V.
   Kart darbe (surge) testinden gecmedi: dayanim iddia edilmez.
   DE hatlarinda 10k asagi cekme: MCU resetteyken surucu kapali kalir.
   RxD, surucu konusurken yuksek empedans: MCU ic pull-up i acilir.
5) EKRAN TEK UCTAN PE ye baglanir, iki uctan baglanirsa ekran uzerinden
   toprak dongusu akimi akar. IEC 61000-4-6 baglikligi duser.
6) MODBUS SALT OKUNUR olacak firmware de yazma/kesme komutu taninmamaz.
   Kesme yetkisi olan bir cihazi aga acmak saldiri yuzeyi olur.
7) ISOPOWER EMISYONU (ADI AN-1349): VISOOUT -> VISOIN ve donusturucu
   topragi (11, 14) -> hat topragi (16, 20) arasinda ferrit. Ferritlerin
   altinda hicbir katmanda bakir dokulmez.

Melike Abigan

Sheet: /07-haberlesme/
File: 07-haberlesme.kicad_sch

Title: Haberlesme — 2 x izole RS-485

Size: A1	Date: 2026-10-11	Rev: C
KiCad E.D.A. 10.0.6		Id: 8/8